# CAPP 30235
# Databases for Public Policy
Spring 2023

**Instructor:** Nicholas Ross

**Contact email:** nickross@uchicago.edu

**Office Hours:** Monday 12-4pm JCL 205 & Wednesday 12-4pm JCL 3C common area

**Data Location:** `https://github.com/NickRoss/sql-data`

**Textbooks:** No textbook is required. The lecture notes for this course, entitled *Data Management* will be covered in the course. Note that since the timing of this course is different than previously taught I'll be releasing the notes after the class date.

In case you want an additional reference or do not like my notes, the following are books and documentation that I have found useful over the years.

- *PostgreSQL Documentation* which can be found at `https://www.postgresql.org/docs/`. This is the PostgreSQL's documentation and is incredibly detailed and easy to follow.
- *Learning SQL* by *Alan Beaulieu* is a great introduction to SQL. It is a bit dated and doesn't cover the advanced topics that we will cover, but if you have never used SQL and want a well-written book, this is a great place to look.
- *SQL for Data Analysis* by *Cathy Tanimura* is a good reference for some of the advanced recipes that we will see in this course.
- *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython* by *Wes McKinney* is the book about Pandas, written by Wes McKinney – the author of Pandas!

# 1 Introduction

A key component of modern data science is extracting and transforming data. The primary focus of this class is learning how to efficiently manipulate data using industry standard tools and within industry standard quality. While the basics of data manipulation and management tend not to be as fun as running models, without a firm grasp of these basic tools working on more complex tasks are much more difficult.

For modern data scientists and engineers, the two most common tools for manipulating data are SQL and Pandas. In this course we will focus on these two. The variant of SQL used in this class, PostgreSQL is popular in the data science field and is, arguably, the most modern of all relational databases systems. The second stream of content is using Python to manipulate data with NumPy and Pandas, two libraries that are used to reshape, manipulate and "wrangle" data.

At the end of this course, I expect students to easily manipulate data within Python and SQL as well as understand the basic trade-offs involved in moving between these tools and how to leverage that knowledge when applying data science techniques.

The challenge of this class is that we are learning a new language and a new way of thinking about data manipulation. Just like learning any new language, the most efficient way to learn is via strategic repetition, which this class will provide plenty of. I expect students to spend 10-20 hours per week outside of class on this course. If you are uncomfortable with that level of commitment, this may not be the best use of your time. This course demands *a lot*, but from it you will gain the skills necessary to be a data scientist.

This course will not just cover the basics of manipulating data, but also the application of those manipulation techniques to common business problems. We will cover concepts such as cohort-analysis, churn prediction, estimating customer lifetime value ("LTV") and other applied techniques in order to dig deeper into the nuances of how databases work.

The course will also cover common interview questions and answers. We will cover the basics of how to prepare for SQL Interviews, how to think about problems and how to answer SQL questions in order to maximize the likelihood of getting that job.

Finally, this course presents an opinionated version of data management. A very opinionated version. In my experiences doing hundreds of data science projects I've seen basic data management be the downfall of far too many. This course leverages that knowledge and experience to put together a practical way of working with data that avoids the most common pitfalls.

## Course Learning Objectives

By the end of this course, you will be able to:

1. Use SQL to extract, transform and manipulate large data sets in preparation for analysis.

2. Be able to use common clauses as SELECT, FROM, WHERE, GROUP BY, and ORDER BY to extract and transform data.

3. Know more advanced analytic/window/partition functions and CTEs and, most importantly when they should and shouldn't be used.

4. Write SQL code using standard industry syntax.

5. Be able to explain and leverage performance assessment techniques to write (and modify) queries to be performant and efficient.

6. Use Pandas to manipulate data within Python. This includes data wrangling, reshaping, manipulating strings, numbers and dates.

7. Write Pandas code which minimizes the likelihood of errors.

8. Efficiently Utilize more advanced Pandas concepts (indexing, multi-indexes, cumulative functions)

9. Understand the trade-offs involved with different data management solutions. Be able to explain and define concepts concerning Relational and NoSQL databases.

10. Utilize common data management techniques to answer applied business focused questions, including churn prediction and estimation of customer LTV.

## Requirements

As a core class within the CAPP Program this course has a number of requirements:

- **Python:** Students should be familiar with Python. This includes:
    1. Loops (for, while)
    2. Conditionals (if)
    3. Data objects (lists and dictionaries)
    4. List comprehensions
    5. Defining functions (def)
    6. Basic File handling (open)
- **General Computing:** Students should have a computer and be able to install Anaconda, Jupyter and Docker on it.
- **Mathematics:** Students should be familiar with:
    1. Calculus I (integration and derivation)
    2. Linear Algebra (manipulation of matrices)

To be clear – the concepts mentioned above are going to be assumed during this class and will be not be explained. If you know you are weak in one of the areas above it would behoove you to refresh your knowledge.

## Assessment

This class uses what is called "continuous assessment" . Under this system, students are provided feedback in a more continuous manner. In particular, this course has quizzes every week, starting in week 2. In terms of overall grades, the breakdown can be found below.

| Type | Percent of Grade |
|---|---|
| Quizes | 15% |
| Homework | 10% |
| Midterm | 25% |
| Final Exam | 40% |
| Professionalism | 10% |
| Total | 100% |

### Quizes

Quizzes are administered at the start of class and will last between 10-20 minutes. Each quiz will ask a few questions about the preceding material and is graded aggressively; the material should be fresh in your mind. Quizzes are done individually and there no make-ups. If a student is late to class they will not be given additional time. Quizzes are done each Thursday and cover the material up to and including the Thursday beforehand.

### Homework Assignments

This class has a ton of homework, all of which is completed in groups. Homework is graded **lightly**. Homework needs to be uploaded by 9:55AM the day that it is due. **Late work is not accepted.**

### Exams

There are two exams in this course: a midterm and a final. Exams are closed-book, closed-notes and no calculator is required. Tentative dates for the exam are below. **All Exams are curved.**

- **Midterm:** 4/20/2023
- **Final Exam:** As scheduled

### Professionalism

In this class professionalism means:

- Showing up on time
- Showing up prepared
- Contributing to class
- Being responsible to your group when doing group work
- Doing required readings
- Communicating respectfully and appropritately to your peers, TAs and Faculty

Students who show up on time, work through the example problems and pay attention in class have a tendency to do extremely well in this class. The opposite if also true; students who fail to make an effort, fail to contribute in class and fail to pay attention have a tendency to do poorly in this class.

## Class Topics and schedule

Since this is the first time that this course has been taught at this University, I'm leaving some leeway in the speed that we will cover material. I'll be publishing the next few weeks of the course on canvas. That being said, here are important dates:

| Date | Event |
|---|---|
| 3/30 | Quiz #1 |
| 4/6 | Quiz #2 |
| 4/13 | Quiz #3 |
| 4/20 | Midterm |
| 5/4 | Quiz #4 |
| 5/11 | Quiz #5 |
| TBD | Final |

## Odds and Ends

- Cheating is not tolerated. At all. Unless an assignment is clearly designated as group work, I expect it to be done alone. The University's code of academic integrity can be found online and I expect it to be followed. Disciplinary action will be taken against any student found violating this code. This includes:
  - Plagiarism – intentionally or unintentionally representing the words or ideas of another person as your own; failure to properly cite references; manufacturing references.
  - Working with another person when independent work is required.
  - Submission of the same paper in more than one course without the specific permission of each instructor.
  - Submitting a paper written by another person or obtained from the internet.
  - The penalties for violation of the policy may include a failing grade on the assignment, a failing grade in the course, and/or a referral to the Academic Integrity Committee.
- Attendance is expected. All exams are in-person and use pencil and paper.
- The quickest way to reach me is via slack.
- Class participation is required. This is a big part of your grade. I expect class to include discussions. Just to make this clear
- If you require any accommodations for this course, as soon as possible please provide the instructor with a copy of your Accommodation Determination Letter (provided to you by the Student Disability Services office) so that you may discuss with them how your accommodations may be implemented in this course. The University of Chicago is committed to ensuring the full participation of all students in its programs. If you have a documented disability (or think you may have a disability) and, as a result, need a reasonable accommodation to participate in class, complete course requirements, or benefit from the University's programs or services, you are encouraged to contact Student Disability Services as soon as possible. To receive reasonable accommodation, you must be appropriately registered with Student Disability Services. Please contact the office at (773) 702-6000, TTY (773) 795-1186, disabilities@uchicago.edu, or visit

the website at disabilities.uchicago.edu. Student Disability Services is located at 5501 S. Ellis Avenue.

- As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I also have a mandatory reporting responsibility related to my role as a faculty member. I am required to share information regarding sexual misconduct or information about a crime that may have occurred with the University.

- The University of Chicago is a community of scholars dedicated to research, academic excellence, and the pursuit and cultivation of learning. Members of the University community cannot thrive unless each is accepted as an autonomous individual and is treated without regard to characteristics irrelevant to participation in the life of the University. Our university is committed to fostering a safe, productive learning environment. Title IX and our school policy prohibits discrimination on the basis of sex. Sexual misconduct including harassment, domestic and dating violence, sexual assault, and stalking is also prohibited at our university.

  Our university encourages anyone experiencing sexual misconduct to talk to someone about what happened, so they can get the support they need and our university can respond appropriately. If you wish to speak confidentially about an incident of sexual misconduct, want more information about filing a report, or have questions about school policies and procedures, please contact Bridget Collier (Associate Provost for Equal Opportunity Programs and Title IX Coordinator for the University) at bcollier@uchicago.edu or (773) 834-6367.

  Our university is legally obligated to investigate reports of sexual misconduct, and therefore it cannot guarantee the confidentiality of a report, but it will consider a request for confidentiality and respect it to the extent possible. As instructors, we are also required by our school to report incidents of sexual misconduct and thus cannot guarantee confidentiality. We must provide our Title IX coordinator with relevant details such as the names of those involved in the incident.